Questions with solutions

This document contains the question bank (50 MCQs) currently used in the quiz, with hints and detailed step-by-step solutions.

Topics Covered:

1. Operators & Special Operators
2. Matrix Algebra
3. Eigenvalues & Eigenvectors
4. Pauli Matrices
5. Commutation Relations

1 Operators & Special Operators

1.1 Question 1

Question

Which mapping is NOT linear (operator on functions f)?

(A) $L(f) = f^2$

(B) $L(f) = \dfrac{df}{dx}$

(C) $L(f) = 3f$

(D) $L(f) = xf(x)$

Hints

- Test $L(f+g) \stackrel{?}{=} L(f) + L(g)$.
- Squaring usually breaks additivity.

Complete Solution

Correct option: (A).
Step 1: Linearity requires $L(f+g) = L(f) + L(g)$ and $L(\alpha f) = \alpha L(f)$ for all f, g and scalars α.
Step 2: For $L(f) = f^2$, we get $L(f+g) = (f+g)^2 = f^2 + 2fg + g^2$.
Step 3: But $L(f) + L(g) = f^2 + g^2$.
Step 4: Since $(f+g)^2 \neq f^2 + g^2$ in general, $L(f) = f^2$ is not linear.

1.2 Question 2

Question

Which statement is always true for any operator A?

(A) $A^\dagger A$ is Hermitian

(B) A is Hermitian

(C) A commutes with every operator

(D) A is invertible

Hints

- Take the dagger: $(A^\dagger A)^\dagger = A^\dagger (A^\dagger)^\dagger$.
- That becomes $A^\dagger A$.

Complete Solution

Correct option: (A).
Step 1: An operator H is Hermitian if $H^\dagger = H$.
Step 2: Use $(XY)^\dagger = Y^\dagger X^\dagger$:

$$(A^\dagger A)^\dagger = A^\dagger (A^\dagger)^\dagger = A^\dagger A.$$

Step 3: Hence $A^\dagger A$ is Hermitian for any A.
Step 4: The other options are not always true for a general operator.

1.3 Question 3

Question

Hermitian condition (matrix/operator H) is:

(A) $H = H^\dagger$

(B) $H^\dagger H = I$

(C) $H^2 = H$

(D) $\det(H) = 1$

Hints

- Hermitian means equal to its conjugate transpose.
- Unitary and projector are different conditions.

Complete Solution

Correct option: (A).
Step 1: By definition, Hermitian (self-adjoint) means $H^\dagger = H$.
Step 2: $H^\dagger H = I$ is the unitary condition, not Hermitian.
Step 3: $H^2 = H$ describes idempotence (projector-like), not Hermiticity.
Step 4: Therefore $H = H^\dagger$ is the defining condition.

1.4 Question 4

Question

Unitary condition (matrix/operator U) is:

(A) $U^\dagger U = I$

(B) $U = U^\dagger$

(C) $U^2 = U$

(D) $\mathrm{tr}(U) = 0$

Hints

- Unitary means inverse equals dagger.
- So $U^\dagger U$ must be identity.

Complete Solution

Correct option: (A).
Step 1: Unitary means $U^{-1} = U^\dagger$.
Step 2: Multiply $U^{-1} = U^\dagger$ on the right by U:

$$U^\dagger U = I.$$

Step 3: Equivalently, $UU^\dagger = I$.
Step 4: Hence the unitary condition is $U^\dagger U = I$.

1.5 Question 5

Question

A projection operator P (in QM sense) must satisfy:

(A) $P^2 = P$ and $P = P^\dagger$

(B) $P^2 = I$ and $P = P^\dagger$

(C) $P^\dagger P = I$

(D) $P = 0$ only

Hints

- Project twice = project once.
- Projectors are Hermitian in QM.

Complete Solution

Correct option: (A).
Step 1: Projection twice should not change the result: $P^2 = P$ (idempotent).
Step 2: In QM, projectors correspond to observables, so they must be Hermitian: $P^\dagger = P$.
Step 3: Option (B) describes an involution, not a projector.
Step 4: Therefore $P^2 = P$ and $P = P^\dagger$.

1.6 Question 6

Question

Let $P = |u\rangle \langle u|$. When is P a projection operator?

(A) When $\langle u|u\rangle = 1$

(B) When $\langle u|u\rangle = 0$

(C) Always, for any $|u\rangle$

(D) Only if $|u\rangle$ is an eigenvector of every operator

Hints

- Check $P^2 = |u\rangle \langle u| |u\rangle \langle u|$.
- You need $\langle u|u\rangle = 1$ for idempotence.

Complete Solution

Correct option: (A).
Step 1: Compute:

$$P^2 = (|u\rangle \langle u|)(|u\rangle \langle u|) = |u\rangle \underbrace{\langle u|u\rangle}_{\text{scalar}} \langle u| = \langle u|u\rangle\ P.$$

Step 2: For a projector we need $P^2 = P$.
Step 3: Thus $\langle u|u\rangle\ P = P$ which (for nonzero P) requires $\langle u|u\rangle = 1$.
Step 4: Hence $|u\rangle$ must be normalized.

1.7 Question 7

Question

If P and Q are projection operators and $PQ = QP$, then which is also a projection operator?

(A) PQ

(B) $P + Q$ (always)

(C) $P - Q$ (always)

(D) $P + Q - I$ (always)

Hints

- Check $(PQ)^2$ when $P^2 = P$ and $Q^2 = Q$.
- Commuting matters.

Complete Solution

Correct option: (A).
Step 1: Since P, Q are projectors: $P^2 = P$, $Q^2 = Q$, and $P^\dagger = P$, $Q^\dagger = Q$.
Step 2: Idempotence:

$$(PQ)^2 = PQPQ = P(QP)Q = P(PQ)Q = (P^2)(Q^2) = PQ,$$

using $PQ = QP$.
Step 3: Hermitian:

$$(PQ)^\dagger = Q^\dagger P^\dagger = QP = PQ.$$

Step 4: Therefore PQ is a projection operator.

1.8 Question 8

Question

If A and B are Hermitian, then the commutator $[A, B]$ is:

(A) Anti-Hermitian (skew-Hermitian)

(B) Hermitian

(C) Unitary

(D) A projection

Hints

- Compute $(AB - BA)^\dagger$.
- You get a minus sign.

Complete Solution

Correct option: (A).
Step 1: $[A, B] = AB - BA$.
Step 2: Take dagger and use $(XY)^\dagger = Y^\dagger X^\dagger$ and $A^\dagger = A$, $B^\dagger = B$:

$$([A, B])^\dagger = (AB - BA)^\dagger = B^\dagger A^\dagger - A^\dagger B^\dagger = BA - AB.$$

Step 3: So $([A, B])^\dagger = -(AB - BA) = -[A, B]$.
Step 4: This is the definition of anti-Hermitian.

1.9 Question 9

Question

If U is unitary and $U|\chi\rangle = \lambda|\chi\rangle$, then:

(A) $|\lambda| = 1$

(B) λ must be real

(C) λ must be 0

(D) λ must be ± 2

Hints

- Unitary preserves norm: $\|U|\chi\rangle\| = \||\chi\rangle\|$.
- So $|\lambda|$ must be 1.

Complete Solution

Correct option: (A).
Step 1: Unitary means $U^\dagger U = I$.
Step 2: Compute the norm:

$$\|U|\chi\rangle\|^2 = \langle\chi|U^\dagger U|\chi\rangle = \langle\chi|\chi\rangle.$$

Step 3: But $U|\chi\rangle = \lambda|\chi\rangle$ gives:

$$\|U|\chi\rangle\|^2 = \|\lambda|\chi\rangle\|^2 = |\lambda|^2\langle\chi|\chi\rangle.$$

Step 4: Equate both: $|\lambda|^2\langle\chi|\chi\rangle = \langle\chi|\chi\rangle$, hence $|\lambda| = 1$ (for nonzero $|\chi\rangle$).

1.10 Question 10

Question

If $[A, B] = 0$, then which is true?

(A) $[f(A), B] = 0$ for any polynomial f

(B) $[A, B] = I$

(C) A and B must be equal

(D) A must be unitary

Hints

- If $AB = BA$, then $A^n B = BA^n$.
- So any polynomial in A also commutes with B.

Complete Solution

Correct option: (A).
Step 1: $[A, B] = 0 \Rightarrow AB = BA$.
Step 2: Then by induction, $A^n B = BA^n$ for all integers $n \geq 1$.
Step 3: For a polynomial $f(A) = \sum_{n=0}^{N} c_n A^n$, we have

$$f(A)B = \sum c_n A^n B = \sum c_n BA^n = Bf(A).$$

Step 4: Hence $[f(A), B] = 0$.

2 Matrix Algebra

2.1 Question 11

Question

For a 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $\det(A)$ equals:

(A) $ad - bc$

(B) $ab - cd$

(C) $a + d$

(D) $b + c$

Hints

- Use the standard 2×2 determinant formula.
- Diagonal product minus off-diagonal product.

Complete Solution

Correct option: (A).
Step 1: For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

$$\det(A) = a \cdot d - b \cdot c.$$

Step 2: Substitute and simplify (already in simplest form).
Step 3: Therefore $\det(A) = ad - bc$.

2.2 Question 12

Question

Which identity is always true (same-size matrices A, B)?

(A) $\mathrm{tr}(AB) = \mathrm{tr}(BA)$

(B) $\det(A + B) = \det(A) + \det(B)$

(C) $(A + B)^2 = A^2 + B^2$

(D) $AB = BA$ always

Hints

- Trace has cyclic property for products.
- Commutativity of matrices is not guaranteed.

Complete Solution

Correct option: (A).
Step 1: Trace satisfies cyclicity: $\mathrm{tr}(XY) = \mathrm{tr}(YX)$ for compatible matrices.
Step 2: Set $X = A$ and $Y = B$ to get $\mathrm{tr}(AB) = \mathrm{tr}(BA)$.
Step 3: The other identities fail in general unless extra conditions hold.
Step 4: Hence (A) is always true.

2.3 Question 13

Question

Transpose of a product satisfies:

(A) $(AB)^T = B^T A^T$

(B) $(AB)^T = A^T B^T$

(C) $(AB)^T = AB$

(D) $(AB)^T = A + B$

Hints

- Order reverses under transpose.
- Try a 2×2 example to verify.

Complete Solution

Correct option: (A).
Step 1: The transpose reverses order: $(XY)^T = Y^T X^T$.
Step 2: Apply with $X = A$, $Y = B$.
Step 3: Hence $(AB)^T = B^T A^T$.

2.4 Question 14

Question

Conjugate-transpose (dagger) of a product satisfies:

(A) $(AB)^\dagger = B^\dagger A^\dagger$

(B) $(AB)^\dagger = A^\dagger B^\dagger$

(C) $(AB)^\dagger = AB$

(D) $(AB)^\dagger = A + B$

Hints

- Dagger reverses order like transpose.
- Also complex-conjugates entries.

Complete Solution

Correct option: (A).
Step 1: Adjoint satisfies $(XY)^\dagger = Y^\dagger X^\dagger$.
Step 2: Put $X = A$, $Y = B$.
Step 3: Therefore $(AB)^\dagger = B^\dagger A^\dagger$.

2.5 Question 15

Question

Find x such that $\det \begin{pmatrix} x & 1 \\ 2 & 3 \end{pmatrix} = 0$.

(A) $x = \frac{2}{3}$

(B) $x = \frac{3}{2}$

(C) $x = -\frac{2}{3}$

(D) $x = 0$

Hints

- $\det = 3x - 2$.
- Set $3x - 2 = 0$.

Complete Solution

Correct option: (A).
Step 1: Compute determinant:

$$\det \begin{pmatrix} x & 1 \\ 2 & 3 \end{pmatrix} = 3x - 2.$$

Step 2: Set to zero: $3x - 2 = 0$.
Step 3: Solve: $x = \frac{2}{3}$.

2.6 Question 16

Question

If Q is an orthogonal matrix (real), then:

(A) $Q^{-1} = Q^T$

(B) $Q^{-1} = Q$

(C) $Q^T Q = 0$

(D) $\det(Q) = 0$ always

Hints

- Orthogonal means $Q^T Q = I$.
- So multiply both sides appropriately.

Complete Solution

Correct option: (A).
Step 1: Orthogonal means $Q^T Q = I$.
Step 2: By definition, the inverse satisfies $Q^{-1} Q = I$.
Step 3: Comparing, we identify $Q^{-1} = Q^T$.

2.7 Question 17

Question

If A is invertible, which statement is true?

(A) $\det(A) \neq 0$

(B) $\mathrm{tr}(A) \neq 0$

(C) A must be diagonal

(D) A must be symmetric

Hints

- Invertibility is tied to determinant for square matrices.
- Trace can be zero even if invertible.

Complete Solution

Correct option: (A).
Step 1: A square matrix is invertible iff its determinant is nonzero.
Step 2: If $\det(A) = 0$, the matrix compresses volume to zero and cannot be inverted.
Step 3: Hence invertible $\Rightarrow \det(A) \neq 0$.

2.8 Question 18

Question

Which is always true for identity matrix I?

(A) $AI = IA = A$

(B) $AI = I$

(C) $IA = 0$

(D) $AI = A^2$

Hints

- Identity behaves like multiplying by 1.
- It should not change A.

Complete Solution

Correct option: (A).
Step 1: By definition of identity, $Iv = v$ for all vectors v.
Step 2: Therefore multiplying any matrix by I leaves it unchanged: $AI = A$ and $IA = A$.
Step 3: So $AI = IA = A$.

2.9 Question 19

Question

If U is unitary, which statement is always true?

(A) $U^{-1} = U^{\dagger}$

(B) $U = U^{\dagger}$

(C) $U^2 = U$

(D) $\det(U) = 0$

Hints

- Definition: $U^{\dagger}U = I$.
- Compare with inverse definition.

Complete Solution

Correct option: (A).
Step 1: Unitary means $U^{\dagger}U = I$.
Step 2: This shows $U^{\dagger}$ is the inverse of U.
Step 3: Therefore $U^{-1} = U^{\dagger}$.

2.10 Question 20

Question

Trace under similarity transform: for invertible S, $\mathrm{tr}(S^{-1}AS)$ equals:

(A) $\mathrm{tr}(A)$

(B) $\det(A)$

(C) $\mathrm{rank}(A)$

(D) 0 always

Hints

- Trace is invariant under similarity transforms.
- Try A diagonal and S change-of-basis.

Complete Solution

Correct option: (A).
Step 1: Use cyclicity: $\mathrm{tr}(XYZ) = \mathrm{tr}(ZXY)$.
Step 2: Then $\mathrm{tr}(S^{-1}AS) = \mathrm{tr}(ASS^{-1})$.
Step 3: Since $SS^{-1} = I$, $\mathrm{tr}(ASS^{-1}) = \mathrm{tr}(AI) = \mathrm{tr}(A)$.

3 Eigenvalues & Eigenvectors

3.1 Question 21

Question

Eigenvalues λ of A are found from:

(A) $\det(A - \lambda I) = 0$

(B) $\mathrm{tr}(A - \lambda I) = 0$

(C) $A - \lambda I = 0$ (always)

(D) $\det(A) = 0$ (always)

Hints

- Characteristic polynomial comes from determinant.
- Nontrivial solutions require determinant zero.

Complete Solution

Correct option: (A).
Step 1: Eigenvalue equation: $Av = \lambda v$ for some $v \neq 0$.
Step 2: Rearranging: $(A - \lambda I)v = 0$.
Step 3: Nonzero solutions exist iff $\det(A - \lambda I) = 0$.
Step 4: Hence eigenvalues satisfy $\det(A - \lambda I) = 0$.

3.2 Question 22

Question

For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, eigenvalues are:

(A) 1 and 3

(B) 0 and 4

(C) 2 and 2

(D) -1 and 5

Hints

- $\det \begin{pmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{pmatrix} = (2-\lambda)^2 - 1$.
- Set to 0.

Complete Solution

Correct option: (A).
Step 1: Compute characteristic determinant:

$$\det(A - \lambda I) = (2-\lambda)^2 - 1.$$

Step 2: Set to zero: $(2-\lambda)^2 - 1 = 0 \Rightarrow (2-\lambda)^2 = 1$.
Step 3: So $2 - \lambda = \pm 1$, giving $\lambda = 1, 3$.

3.3 Question 23

Question

For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, an eigenvector for $\lambda = 3$ is proportional to:

(A) $(1, 1)$

(B) $(1, -1)$

(C) $(2, 0)$

(D) $(0, 2)$

Hints

- Solve $(A - 3I)v = 0$.
- You get $v_1 = v_2$.

Complete Solution

Correct option: (A).
Step 1: $A - 3I = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$.
Step 2: Solve $\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \binom{v_1}{v_2} = 0$, giving $-v_1 + v_2 = 0$.
Step 3: So $v_2 = v_1$, hence $v \propto (1, 1)$.

3.4 Question 24

Question

For diagonal matrix $\text{diag}(2, 5, -1)$, the eigenvalues are:

(A) $2, 5, -1$

(B) Only 2

(C) Only 5

(D) $2 + 5 - 1 = 6$

Hints

- Diagonal matrices scale basis vectors.
- Each diagonal entry is an eigenvalue.

Complete Solution

Correct option: (A).
Step 1: For a diagonal matrix, $Ae_i = a_{ii}e_i$.
Step 2: Thus each diagonal entry is an eigenvalue.
Step 3: Hence eigenvalues are $2, 5, -1$.

3.5 Question 25

Question

If $Av = \lambda v$, then $A^2 v$ equals:

(A) $\lambda^2 v$

(B) $2\lambda v$

(C) λv

(D) v/λ

Hints

- Apply A again to Av.
- $A(Av) = A(\lambda v) = \lambda(Av) = \lambda(\lambda v)$.

Complete Solution

Correct option: (A).
Step 1: Start with $Av = \lambda v$.
Step 2: Apply A again: $A^2 v = A(Av) = A(\lambda v)$.
Step 3: Pull out scalar: $A(\lambda v) = \lambda Av = \lambda(\lambda v) = \lambda^2 v$.

3.6 Question 26

Question

If A is invertible and $Av = \lambda v$ ($\lambda \neq 0$), then $A^{-1}v$ equals:

(A) $(1/\lambda)\, v$

(B) λv

(C) 0

(D) Undefined

Hints

- Multiply $Av = \lambda v$ by A^{-1} on the left.
- You get $v = \lambda A^{-1} v$.

Complete Solution

Correct option: (A).
Step 1: Multiply $Av = \lambda v$ by A^{-1}:

$$A^{-1}Av = A^{-1}(\lambda v).$$

Step 2: Simplify: $v = \lambda A^{-1} v$.
Step 3: Divide by λ (since $\lambda \neq 0$): $A^{-1}v = (1/\lambda)\, v$.

3.7 Question 27

Question

For any square matrix A, the sum of eigenvalues (with multiplicity) equals:

(A) $\mathrm{tr}(A)$

(B) $\det(A)$

(C) $\mathrm{rank}(A)$

(D) 0 always

Hints

- Check for diagonal matrices first.
- Trace matches sum of diagonal entries.

Complete Solution

Correct option: (A).
Step 1: For diagonal A, eigenvalues are diagonal entries, and $\mathrm{tr}(A)$ is their sum.
Step 2: For general A, trace is invariant under similarity transforms, and equals the sum of eigenvalues (counting multiplicity).
Step 3: Hence sum of eigenvalues $= \mathrm{tr}(A)$.

3.8 Question 28

Question

For any square matrix A, the product of eigenvalues (with multiplicity) equals:

(A) $\det(A)$

(B) $\mathrm{tr}(A)$

(C) $\mathrm{rank}(A)$

(D) 1 always

Hints

- Check for diagonal matrices first.
- Determinant multiplies diagonal entries.

Complete Solution

Correct option: (A).
Step 1: For diagonal A, $\det(A)$ is the product of diagonal entries (the eigenvalues).
Step 2: In general, determinant equals the product of eigenvalues (counting multiplicity).
Step 3: Therefore product of eigenvalues $= \det(A)$.

3.9 Question 29

Question

For Hermitian operator H, eigenvectors corresponding to distinct eigenvalues are:

(A) Orthogonal

(B) Always equal

(C) Always non-orthogonal

(D) Always complex

Hints

- This is a key theorem for Hermitian operators.
- It underpins orthonormal eigenbases.

Complete Solution

Correct option: (A).
Step 1: Let $Hv_1 = \lambda_1 v_1$ and $Hv_2 = \lambda_2 v_2$ with $\lambda_1 \neq \lambda_2$.
Step 2: Compute $\langle v_1 | Hv_2 \rangle = \lambda_2 \langle v_1 | v_2 \rangle$.
Step 3: Using $H^\dagger = H$, also $\langle v_1 | Hv_2 \rangle = \langle Hv_1 | v_2 \rangle = \lambda_1 \langle v_1 | v_2 \rangle$.
Step 4: Thus $(\lambda_1 - \lambda_2) \langle v_1 | v_2 \rangle = 0$, so $\langle v_1 | v_2 \rangle = 0$.

3.10 Question 30

Question

If $[A, B] = 0$ and A is Hermitian with non-degenerate eigenvalues, then:

(A) A and B can share eigenvectors (simultaneous eigenbasis)

(B) A and B must be equal

(C) B must be a projection

(D) A cannot be diagonalized

Hints

- Commuting + non-degenerate implies shared eigenvectors.
- This is the standard compatibility statement.

Complete Solution

Correct option: (A).
Step 1: $[A, B] = 0$ implies $AB = BA$.
Step 2: If $Av = \lambda v$, then $A(Bv) = B(Av) = \lambda(Bv)$, so Bv lies in the same eigenspace of A.
Step 3: If λ is non-degenerate, that eigenspace is 1D, so Bv must be proportional to v.
Step 4: Hence v is a simultaneous eigenvector, giving a common eigenbasis.

4 Pauli Matrices

4.1 Question 31

Question

Which matrix is σ_x?

(A) $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

(B) $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

(C) $\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$

(D) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

Hints

- σ_x swaps basis states.
- It has ones off-diagonal.

Complete Solution

Correct option: (A).
Step 1: The Pauli matrices are standard:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Step 2: Hence σ_x is option (A).

4.2 Question 32

Question

Which matrix is σ_y?

(A) $\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$

(B) $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

(C) $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

(D) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

Hints

- σ_y has imaginary off-diagonals.
- It is Hermitian.

Complete Solution

Correct option: (A).
Step 1: By definition,

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$

Step 2: So the correct option is (A).

4.3 Question 33

Question

Which matrix is σ_z?

(A) $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

(B) $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

(C) $\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$

(D) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

Hints

- σ_z is diagonal.
- Entries are $+1$ and -1.

Complete Solution

Correct option: (A).
Step 1: By definition,

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Step 2: So the correct option is (A).

4.4 Question 34

Question

Eigenvalues of σ_z are:

(A) $+1$ and -1

(B) 0 and 1

(C) i and $-i$

(D) 2 and -2

Hints

- σ_z is diagonal.
- Diagonal entries are eigenvalues.

Complete Solution

Correct option: (A).
Step 1: $\sigma_z = \mathrm{diag}(1, -1)$ is diagonal.
Step 2: Therefore eigenvalues are the diagonal entries: 1 and -1.

4.5 Question 35

Question

Which product identity is correct?

(A) $\sigma_x\sigma_y = i\sigma_z$

(B) $\sigma_x\sigma_y = -i\sigma_z$

(C) $\sigma_x\sigma_y = \sigma_z$

(D) $\sigma_x\sigma_y = iI$

Hints

- Standard Pauli multiplication.
- Order matters: $\sigma_y\sigma_x = -i\sigma_z$.

Complete Solution

Correct option: (A).
Step 1: Pauli matrices satisfy $\sigma_i\sigma_j = \delta_{ij}I + i\varepsilon_{ijk}\sigma_k$.
Step 2: For $i = x$, $j = y$: $\delta_{xy} = 0$ and $\varepsilon_{xyz} = +1$.
Step 3: Therefore $\sigma_x\sigma_y = i\sigma_z$.

4.6 Question 36

Question

The commutator $[\sigma_x, \sigma_y]$ equals:

(A) $2i\sigma_z$

(B) $i\sigma_z$

(C) $2\sigma_z$

(D) 0

Hints

- Use $[\sigma_i, \sigma_j] = 2i\,\varepsilon_{ijk}\sigma_k$.
- So x, y gives z.

Complete Solution

Correct option: (A).
Step 1: Use the identity $[\sigma_i, \sigma_j] = 2i\,\varepsilon_{ijk}\sigma_k$.
Step 2: For $(i, j) = (x, y)$, $\varepsilon_{xyz} = +1$.
Step 3: Hence $[\sigma_x, \sigma_y] = 2i\sigma_z$.

4.7 Question 37

Question

For $i \neq j$, the anti-commutator $\{\sigma_i, \sigma_j\}$ equals:

(A) 0

(B) $2I$

(C) $2\sigma_k$

(D) I

Hints

- Distinct Pauli matrices anticommute.
- $\sigma_x\sigma_y = -\sigma_y\sigma_x$.

Complete Solution

Correct option: (A).
Step 1: Pauli anti-commutator identity: $\{\sigma_i, \sigma_j\} = 2\delta_{ij}I$.
Step 2: If $i \neq j$, then $\delta_{ij} = 0$.
Step 3: Hence $\{\sigma_i, \sigma_j\} = 0$.

4.8 Question 38

Question

Which identity is always true?

(A) $\sigma_i^2 = I$

(B) $\sigma_i^2 = 0$

(C) $\sigma_i^2 = \sigma_i$

(D) $\sigma_i^2 = -I$

Hints

- Check $\sigma_x \sigma_x$.
- It becomes identity.

Complete Solution

Correct option: (A).
Step 1: Using $\sigma_i \sigma_j = \delta_{ij} I + i\varepsilon_{ijk}\sigma_k$.
Step 2: Set $i = j$ to get $\sigma_i^2 = I$ since $\varepsilon_{iik} = 0$.
Step 3: Therefore $\sigma_i^2 = I$ for $i = x, y, z$.

4.9 Question 39

Question

Trace of $\sigma_x, \sigma_y, \sigma_z$ is:

(A) 0

(B) 1

(C) 2

(D) -1

Hints

- Trace is sum of diagonal entries.
- σ_x and σ_y have 0 diagonals; σ_z has $1 + (-1)$.

Complete Solution

Correct option: (A).
Step 1: $\mathrm{tr}(\sigma_x) = 0 + 0 = 0.$
Step 2: $\mathrm{tr}(\sigma_y) = 0 + 0 = 0.$
Step 3: $\mathrm{tr}(\sigma_z) = 1 + (-1) = 0.$
Step 4: Hence trace of any Pauli matrix is 0.

4.10 Question 40

Question

Determinant of any Pauli matrix is:

(A) -1

(B) $+1$

(C) 0

(D) Depends on i

Hints

- Compute $\det(\sigma_z) = \det \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = -1.$
- Same for others.

Complete Solution

Correct option: (A).
Step 1: For $\sigma_z = \mathrm{diag}(1, -1)$, $\det(\sigma_z) = -1$.
Step 2: $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ has determinant $0 \cdot 0 - 1 \cdot 1 = -1$.
Step 3: $\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ has determinant $0 \cdot 0 - (-i)(i) = -1$.
Step 4: Hence all Pauli matrices have determinant -1.

5 Commutation Relations

5.1 Question 41

Question

The commutator $[A, B]$ is defined as:

(A) $AB - BA$

(B) $AB + BA$

(C) $A + B$

(D) $A - B$

Hints

- Commutator measures non-commutativity.
- If $AB = BA$, commutator is zero.

Complete Solution

Correct option: (A).
Step 1: By definition, $[A, B] = AB - BA$.
Step 2: If $AB = BA$, then $[A, B] = 0$ (they commute).

5.2 Question 42

Question

Which commutator is always zero for any operator A?

(A) $[A, I]$

(B) $[A, A^\dagger]$

(C) $[A, B]$ for any B

(D) $[A, A^\dagger A]$

Hints

- Identity commutes with everything.
- $AI = IA$.

Complete Solution

Correct option: (A).
Step 1: $[A, I] = AI - IA$.
Step 2: Since $AI = IA = A$, we get $AI - IA = 0$.
Step 3: Hence $[A, I] = 0$ always.

5.3 Question 43

Question

If $[A, B] = 0$, then:

(A) $AB = BA$

(B) $AB = 0$

(C) $A = B$

(D) A must be Hermitian

Hints

- Rewrite the definition.
- $[A, B] = AB - BA$.

Complete Solution

Correct option: (A).
Step 1: $[A, B] = 0 \Rightarrow AB - BA = 0$.
Step 2: Rearranging gives $AB = BA$.

5.4 Question 44

Question

Product rule for commutators is:

(A) $[A, BC] = [A, B]C + B[A, C]$

(B) $[A, BC] = [A, B] + [A, C]$

(C) $[A, BC] = [A, B][A, C]$

(D) $[A, BC] = 0$ always

Hints

- Expand $A(BC) - (BC)A$.
- Then regroup terms.

Complete Solution

Correct option: (A).
Step 1: Start from definition:

$$[A, BC] = ABC - BCA.$$

Step 2: Add and subtract BAC:

$$ABC - BCA = (ABC - BAC) + (BAC - BCA).$$

Step 3: Factor:

$$(ABC - BAC) = (AB - BA)C = [A, B]C, \quad (BAC - BCA) = B(AC - CA) = B[A, C].$$

Step 4: Hence $[A, BC] = [A, B]C + B[A, C]$.

5.5 Question 45

Question

Jacobi identity is:

(A) $[A, [B, C]] + [B, [C, A]] + [C, [A, B]] = 0$

(B) $[A, B] + [B, C] + [C, A] = 0$

(C) $[A, B] = [B, A]$

(D) $[A, [A, B]] = 0$ always

Hints

- This is the standard triple-commutator identity.
- It sums to zero.

Complete Solution

Correct option: (A).
Step 1: The Jacobi identity for commutators is

$$[A, [B, C]] + [B, [C, A]] + [C, [A, B]] = 0.$$

Step 2: It is a defining property of Lie algebras and ensures consistency of the bracket operation.

5.6 Question 46

Question

Canonical commutator (1D) is:

(A) $[x, p] = i\hbar$

(B) $[x, p] = 0$

(C) $[x, p] = \hbar$

(D) $[x, p] = -\hbar$

Hints

- This is a postulate of QM.
- Position and momentum do not commute.

Complete Solution

Correct option: (A).
Step 1: In 1D quantum mechanics, the fundamental canonical commutator is

$$[x, p] = i\hbar.$$

Step 2: This non-commutativity leads to the Heisenberg uncertainty principle.

5.7 Question 47

Question

Given $[X, P] = i\hbar$, what is $[X, P^2]$?

(A) $2i\hbar\, P$

(B) $i\hbar\, P^2$

(C) 0

(D) $-2i\hbar\, P$

Hints

- Use $[A, BC] = [A, B]C + B[A, C]$.
- So $[X, P^2] = [X, P]P + P[X, P]$.

Complete Solution

Correct option: (A).
Step 1: Use the product rule:

$$[X, P^2] = [X, PP] = [X, P]P + P[X, P].$$

Step 2: Substitute $[X, P] = i\hbar$:

$$[X, P^2] = (i\hbar)P + P(i\hbar).$$

Step 3: Since $i\hbar$ is a scalar, it commutes with P, giving $2i\hbar P$.

5.8 Question 48

Question

If A and B are Hermitian and AB is also Hermitian, then:

(A) $[A, B] = 0$

(B) $A = B$

(C) A must be unitary

(D) B must be a projector

Hints

- $(AB)^\dagger = B^\dagger A^\dagger = BA$.
- If AB is Hermitian, $AB = (AB)^\dagger$.

Complete Solution

Correct option: (A).
Step 1: If AB is Hermitian, then $AB = (AB)^\dagger$.
Step 2: Compute $(AB)^\dagger = B^\dagger A^\dagger = BA$ since $A^\dagger = A$ and $B^\dagger = B$.
Step 3: Hence $AB = BA$, so $[A, B] = 0$.

5.9 Question 49

Question

If two observables commute and A is non-degenerate, then they can be:

(A) Simultaneously diagonalized (common eigenbasis)

(B) Measured only one at a time

(C) Equal to identity

(D) Only anti-Hermitian

Hints

- Commuting + non-degenerate implies shared eigenvectors.
- This is compatibility in QM.

Complete Solution

Correct option: (A).
Step 1: If $[A, B] = 0$, then B maps each eigenspace of A to itself.
Step 2: If A is non-degenerate, each eigenspace is 1D, so eigenvectors of A are also eigenvectors of B.
Step 3: Therefore they share a common eigenbasis and are simultaneously diagonalizable.

5.10 Question 50

Question

If A and B commute, then which is true?

(A) $[A, B^n] = 0$ for all integers $n \geq 1$

(B) $[A, B] = I$

(C) $[A, B] = 2AB$

(D) A must be a projection

Hints

- If $AB = BA$ then $AB^n = B^n A$.
- Induction works.

Complete Solution

Correct option: (A).
Step 1: Given $AB = BA$.
Step 2: For $n = 1$, $AB^1 = B^1 A$ is true.
Step 3: Assume $AB^n = B^n A$. Then

$$AB^{n+1} = A(B^n B) = (AB^n)B = (B^n A)B = B^n(AB) = B^n(BA) = B^{n+1}A.$$

Step 4: Hence $AB^n = B^n A$ for all $n \geq 1$, so $[A, B^n] = 0$.